Chapter 7: Behavioral Strategies

ECON 5001
Prof. Paul J. Healy
Based on: *Game Theory* by Giacomo Bonanno

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Where We Left Off

Chapter 4: game trees with **information sets**, solved by SPNE

- Solve the smallest subgames first, work backwards
  - A generalization of backwards induction
- We said if a subgame has **no** NE, then there's **no SPNE**

Chapter 6: **mixed strategies**

- **Nash's Theorem**: every finite game (or subgame) has a (possibly mixed) NE
- So the "no NE in a subgame" problem can be fixed up
- But we have to allow mixing inside a game tree

Chapter 7:

1. Put lotteries and mixing into game trees
2. Find a **simpler** way to mix in trees: behavioral strategies
3. Show that a SPNE always exists
4. See what SPNE still gets wrong

Lotteries in Game Trees

“The Game” 2013

2013 OSU @ Michigan. Urban Meyer vs. Brady Hoke. 1,129 total offensive yards
Was 42-35. With 32 sec left, MI scores TD. Now 42-41.

- Brady Hoke: **Kick** to tie/OT or **Go for 2** to win

- **Kick**: Good 95% of the time, almost surely leads to overtime
  - Overtime: let’s say it’s 50-50 who wins
- **Go for 2**: Model it as a 2×2 subgame
  - Michigan: **Run** or **Pass**
  - OSU: **Stop Run (SR)** or **Stop Pass (SP)**

Outcome Payoffs: $u(\text{Win}) = 100$, $u(\text{Lose}) = 0$ for both teams

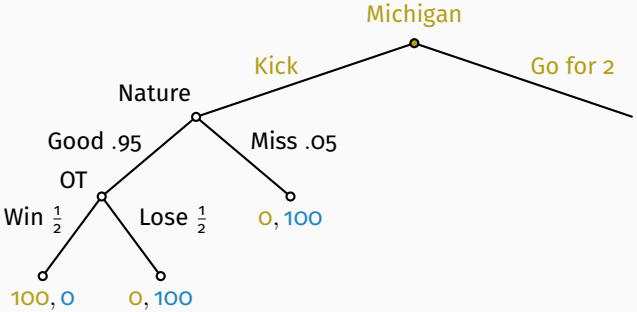
How to draw this tree?

Drawing the Tree



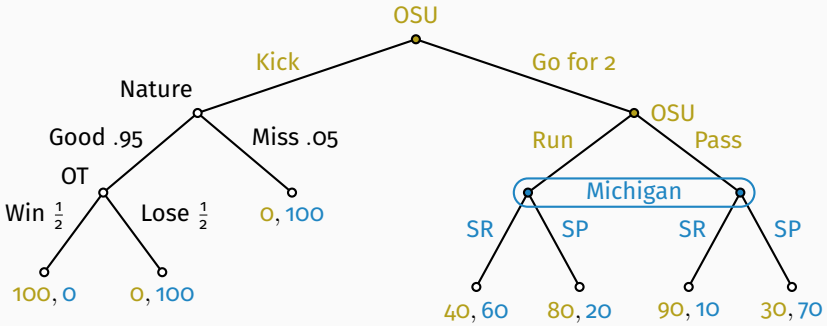
Michigan moves first

Drawing the Tree



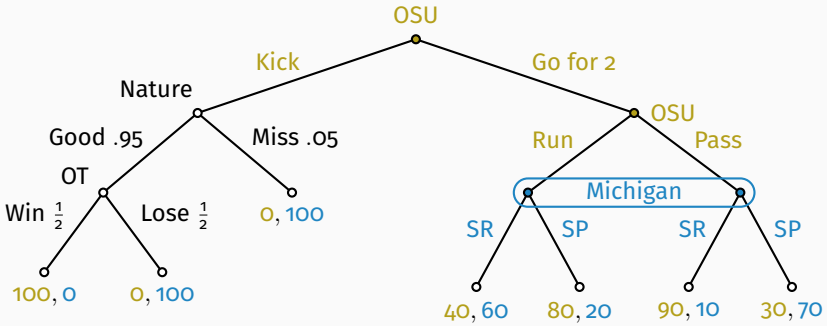
Kick: two moves by “Nature”. Each branch has a probability

Drawing the Tree



Go for 2: a simultaneous-move game hanging off the tree

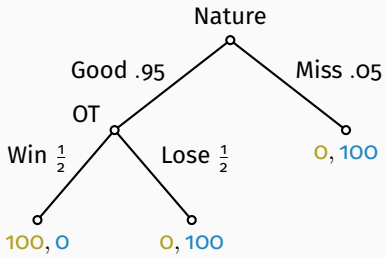
Drawing the Tree



How many proper subgames? (Include moves by Nature.) _____

Nature Gives a Lottery Over Outcomes

After Michigan chooses Kick:



Overtime gives Michigan $\begin{pmatrix} \text{Win} & \text{Lose} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

$U(\text{OT}) =$ _____

Kick gives Michigan $\begin{pmatrix} \text{OT} & \text{Lose} \\ .95 & .05 \end{pmatrix}$

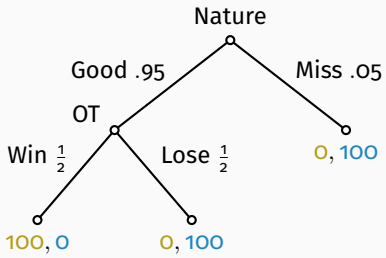
$U(\text{Kick}) =$ _____

Rule: Same as Chapter 6. Use expected utilities

Bonanno (§7.1) writes the lottery as the payoff instead of drawing Nature. Using Nature is more standard.

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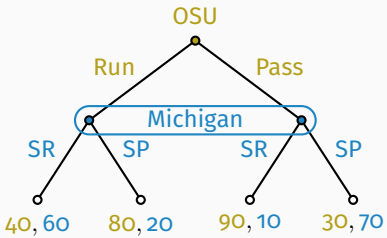
Rule: Same as Chapter 6. Use expected utilities

So: replace the entire Kick branch with _____

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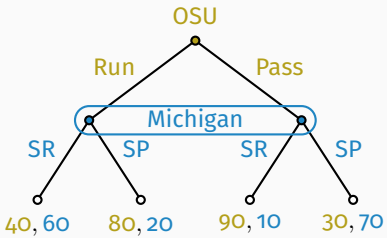
Subgame-Perfect Equilibrium: Allowing Mixing

Step 1: Solve the Subgame



		Michigan	
		Stop Run	Stop Pass
OSU	Run	40, 60	80, 20
	Pass	90, 10	30, 70

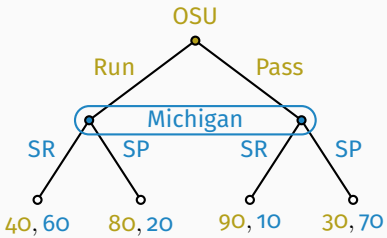
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No pure NE. Chapter 4 (ordinal payoffs): algorithm **halts**. No SPNE!

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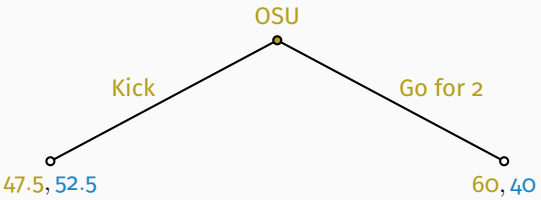


OSU \ Michigan	Stop Run	Stop Pass
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Chapter 6 (cardinal payoffs): mixed NE. OSU Run _____, Michigan Stop Run _____

Payoffs: (_____, _____)

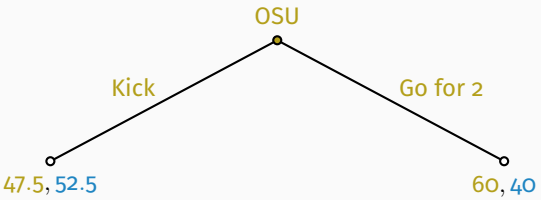
Step 2: Replace and Solve



Kick: Nature's lottery, replaced by its expected utility
Go for 2: the subgame, replaced by its **mixed** NE payoffs

OSU chooses: _____

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Kick: Nature's lottery, replaced by its expected utility
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OSU chooses: _____

Real coaches carry a chart for exactly this decision

The SPNE

Write a probability at **every** decision point:

OSU: $\left(\begin{array}{cc|cc} \text{Kick} & \text{Go} & \text{Run} & \text{Pass} \\ ____ & ____ & ____ & ____ \end{array}\right)$ Michigan: $\left(\begin{array}{cc} \text{SR} & \text{SP} \\ ____ & ____ \end{array}\right)$

The SPNE

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Notice: we didn't write a probability for each **strategy**
We wrote one probability distribution per **information set**

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That's called a **behavioral strategy**

Selten's Theorem

Theorem (Selten, 1965)
Every finite extensive-form game with cardinal payoffs has **at least one** SPNE in mixed strategies

Why?

- Every subgame is a finite game
- Nash: every finite game has a (mixed) NE
- So the SPNE algorithm **never halts**

Ordinal payoffs: SPNE might not exist
Cardinal payoffs: SPNE always exists

Reinhard Selten shared the 1994 Nobel Prize with John Nash and John Harsanyi.

Behavioral Strategies

OSU's Strategies

A strategy: one choice at every information set (Chapter 4)

OSU has two information sets, two choices each:

$$S_{OSU} = \{\ (\text{Kick}, \text{Run}),\ (\text{Kick}, \text{Pass}),\ (\text{Go}, \text{Run}),\ (\text{Go}, \text{Pass})\ \}$$

- (Kick, Run) and (Kick, Pass) are different strategies. Don't combine them!

Mixed strategy (Chapter 6): a probability for each of the 4 strategies

$$\begin{pmatrix} (\text{Kick}, \text{Run}) & (\text{Kick}, \text{Pass}) & (\text{Go}, \text{Run}) & (\text{Go}, \text{Pass}) \\ p & q & r & 1-p-q-r \end{pmatrix}$$

- Needs _____ numbers

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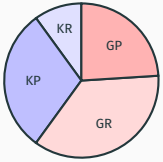
- Needs _____ numbers

Soooo... why would a coach ever plan (Kick, Run)???

Two Ways To Randomize

Mixed strategy

- Before the game: spin one big wheel over **complete plans**
- Then follow the plan



Behavioral strategy

- At each information set, when you get there: flip that set's coin
- One coin per information set



OSU behavioral: $\left(\begin{array}{cc|cc} \text{Kick} & \text{Go} & \text{Run} & \text{Pass} \\ p & 1-p & q & 1-q \end{array}\right)$ Needs _____ numbers

Formal Definition: Behavioral Strategy

Definition (Bonanno 7.1.1)
A **behavioral strategy** for a player is a list of probability distributions, one for every information set of that player. Each one is over the choices at that information set.

How many numbers? Ex: 3 information sets, 2 choices at each

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Fact: Mixed grows by **multiplying**, behavioral grows by **adding**

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Fact: Mixed grows by **multiplying**, behavioral grows by **adding**

Fact: One information set? Then mixed $=$ behavioral (like Michigan)

Are They the Same?

Suppose OSU's coach uses this mixed strategy:

$$\sigma_{OSU} = \begin{pmatrix} (\text{Kick}, \text{Run}) & (\text{Kick}, \text{Pass}) & (\text{Go}, \text{Run}) & (\text{Go}, \text{Pass}) \\ 0.1 & 0.3 & 0.36 & 0.24 \end{pmatrix}$$

Find a behavioral strategy that does the same thing:

$\Pr(\text{Go}) =$ _____

$\Pr(\text{Run} \mid \text{Go}) =$ _____

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Lesson: The 0.1 vs. 0.3 split never matters. Many mixed strategies $\rightarrow$ one behavioral

Kuhn's Theorem

Theorem (Kuhn, 1953)
In games with **perfect recall**, for every mixed strategy there is a behavioral strategy that gives the same probability over terminal nodes

(No matter what the other players do)

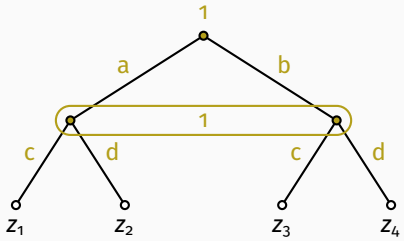
Thus: Use behavioral strategies. Simpler, and nothing is lost

Recall perfect recall (Chapter 4):

- Never forget what you knew
- Never forget what you did

Bonanno works through his own example (Figure 7.3) on pp. 239–241.

Without Perfect Recall



P1 forgets whether they chose a or b

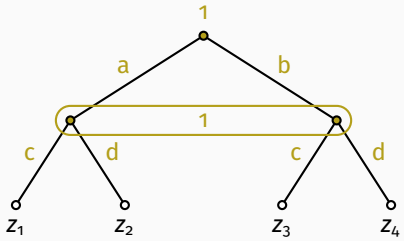
Mixed: $\begin{pmatrix} (a,c) & (a,d) & (b,c) & (b,d) \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{pmatrix}$

Gives z_1: $\frac{1}{2}$, z_4: $\frac{1}{2}$

Behavioral: $\left(\begin{array}{cc|cc} a & b & c & d \\ p & 1-p & q & 1-q \end{array}\right)$

Need $\Pr(z_2) = _____ = 0$

Without Perfect Recall



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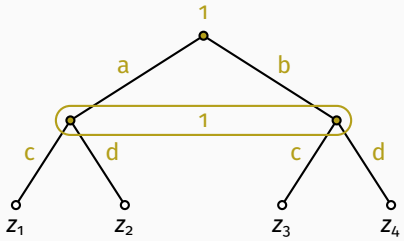
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$\Rightarrow p = 0$ or $q = 1$

$\Rightarrow \Pr(z_1) = 0$ or $\Pr(z_4) = 0$

Without Perfect Recall



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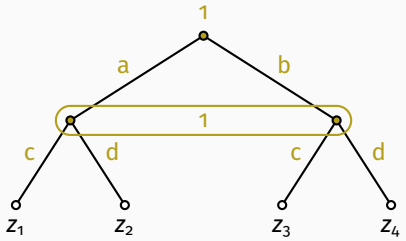
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Can't be done!

Without Perfect Recall



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Can't be done!

Ex: "Player 1" is a team that can't talk once play starts. A pre-game coin flip keeps them coordinated. Separate coin flips can't (not in the book)

Two Ways To Be Unpredictable

Randomize **as you go** (behavioral):

- **Play-calling**: each down, run or pass?
- **Pitching**: each pitch, fastball or curveball?
- **Poker**: each betting round, bluff or fold?
- **Penalty kicks**: each kick, left or right?

Randomize **a whole plan in advance** (mixed):

- **Security**: LAX's ARMOR software draws each day's checkpoint schedule
- **Inspections**: random drug-test lists, tax audits, restaurant inspections
- **Exams**: a random version of the test, handed out before it starts

Kuhn: with perfect recall, the two are equally powerful

Not in the book. ARMOR: Pita et al. (2008, AAMAS).

But Those Are More Complicated

Real versions add:

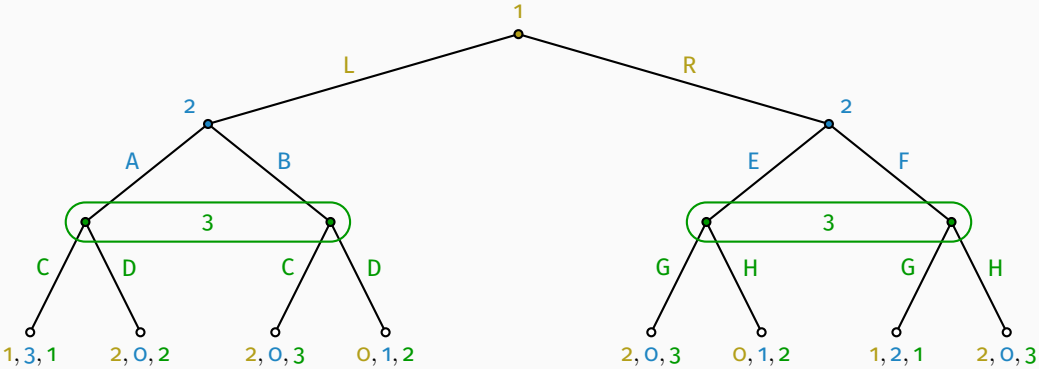
- Many information sets, many choices at each
- Opponents who learn from what you did earlier
- Organizations that **forget**: no perfect recall!
  - New staff, lost notes, no one knows last week's schedule

Core idea survives: one coin at each decision point is enough, **if** you remember

A Bigger Example

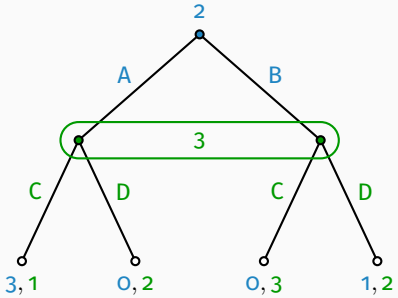
Figure 7.7

Find the SPNE



How many proper subgames? _____ Any pure NE in them?

Figure 7.8: The Left Subgame



Player 2 \ Player 3	C	D
A	3, 1	0, 2
B	0, 3	1, 2

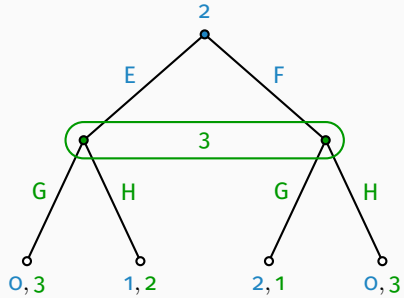
$q = \Pr(C)$ makes P2 indifferent:

$p = \Pr(A)$ makes P3 indifferent:

Payoffs (P1, P2, P3): (_____, _____, _____)

P1's payoffs come from the tree

Figure 7.10: The Right Subgame



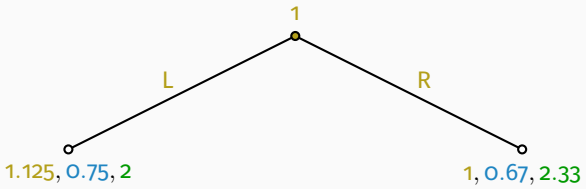
Player 2 \ Player 3	G	H
E	0, 3	1, 2
F	2, 1	0, 3

$q = \Pr(G)$ makes P2 indifferent:

$p = \Pr(E)$ makes P3 indifferent:

Payoffs (P1, P2, P3): (_____, _____, _____)

Figure 7.11: What's Left



Player 1 chooses: _____

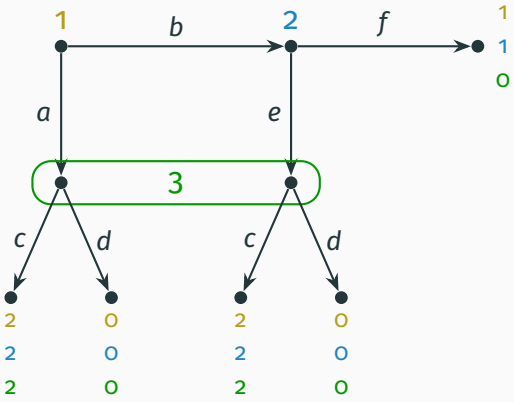
SPNE, in behavioral strategies:

$$\text{P1:} \begin{pmatrix} L & R \\ __ & __ \end{pmatrix} \qquad \text{P2:} \left(\begin{array}{cc|cc} A & B & E & F \\ __ & __ & __ & __ \end{array}\right)$$

$$\text{P3:} \left(\begin{array}{cc|cc} C & D & G & H \\ __ & __ & __ & __ \end{array}\right)$$

What SPNE Gets Wrong

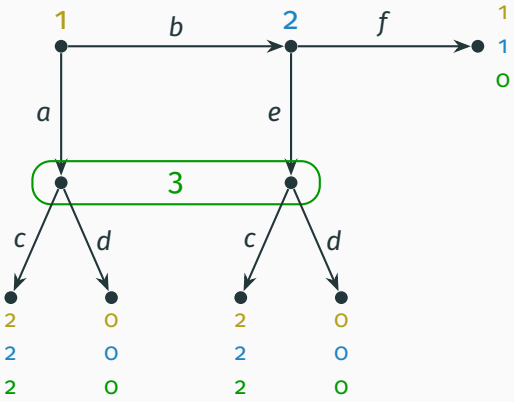
Selten's Horse



Proper subgames? _____

Bonanno Figure 7.12. Named by Selten (1975) for its shape.

Selten's Horse

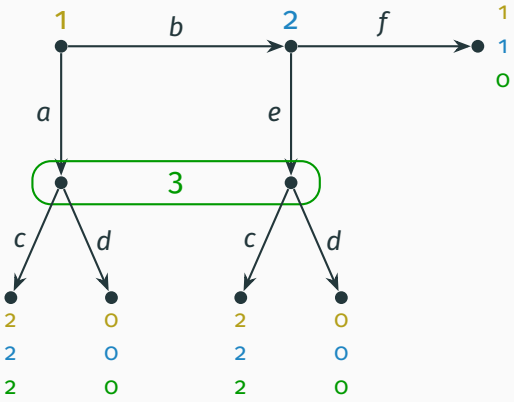


Proper subgames? _____

So SPNE = NE

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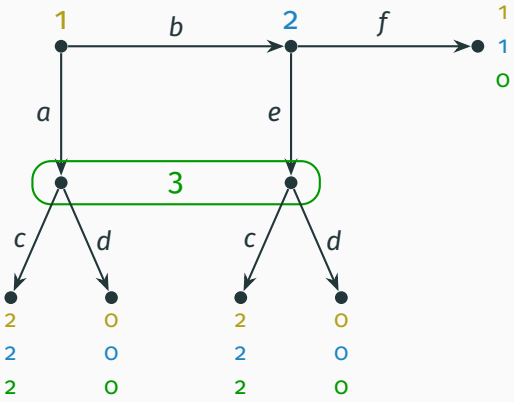
Pure NE:

_____ _____

_____ _____

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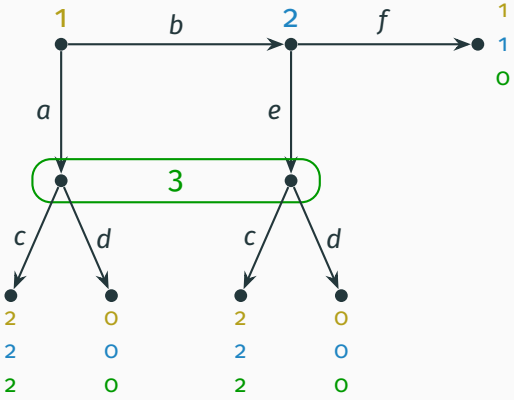
_____ _____

_____ _____

Are they all reasonable???

Bonanno Figure 7.12. Named by Selten (1975) for its shape.

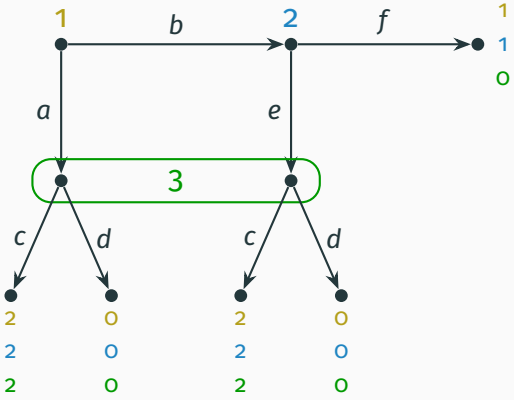
Problem 1: (a, f, c)



P1 plays a. P2 never moves

So P2's plan f costs P2 nothing

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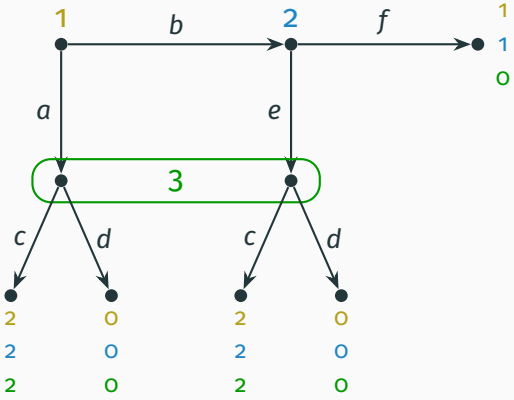
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But **if** P2 had to move?

e gives _____ f gives _____

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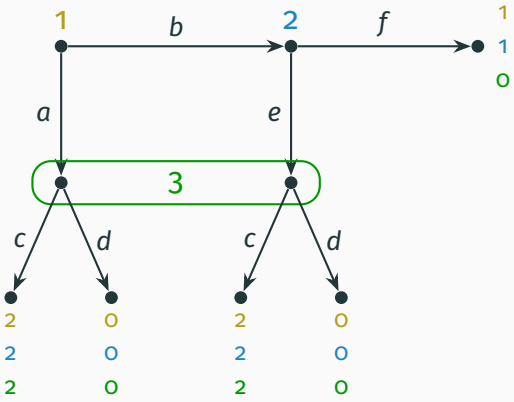
So P2's plan f costs P2 nothing

But **if** P2 had to move?

e gives _____ f gives _____

f is a bad plan. SPNE doesn't notice

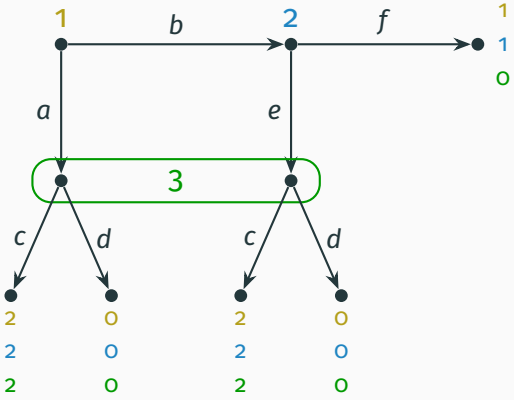
Problem 2: (b, f, d)



P1 plays b, P2 plays f. P3 never moves

Bonanno typo, p. 249: for (b, f, d) it says “Player 1 plays a”. Should be b.

Problem 2: (b, f, d)



P1 plays b, P2 plays f. P3 never moves

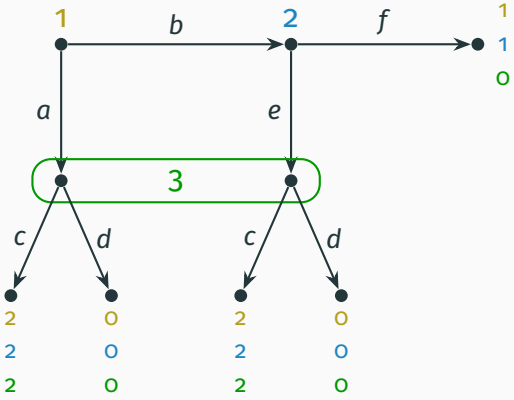
But **if** P3 had to move?

Left node: c _____ vs. d _____

Right node: c _____ vs. d _____

Bonanno typo, p. 249: for (b, f, d) it says "Player 1 plays a". Should be b.

Problem 2: (b, f, d)



P1 plays b, P2 plays f. P3 never moves

But **if** P3 had to move?

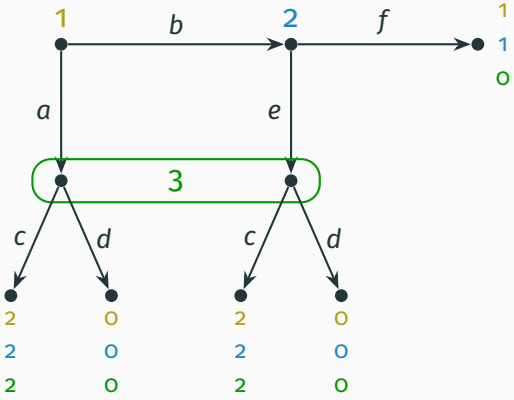
Left node: c _____ vs. d _____

Right node: c _____ vs. d _____

d is **strictly dominated** at P3's infoset!

Bonanno typo, p. 249: for (b, f, d) it says "Player 1 plays a". Should be b.

Problem 2: (b, f, d)



P1 plays b, P2 plays f. P3 never moves

But **if** P3 had to move?

Left node: c _____ vs. d _____

Right node: c _____ vs. d _____

d is **strictly dominated** at P3's infoset!

Lesson: d is dominated as a **choice**, not as a **strategy**
Unreached infosets escape the test

Bonanno typo, p. 249: for (b, f, d) it says "Player 1 plays a". Should be b.

So What's Missing?

SPNE checks for rationality only in **subgames**

- A subgame can't cut an information set
- So choices at big infosets are only checked if they're on the path of play

The fix: Every choice should be sensible, **even off the path of play**

- P3 doesn't know which node she's at
  $\Rightarrow$ she needs a **belief** about where she is

Thus: $\{\text{better equilibria}\} \subseteq SPNE \subseteq NE$

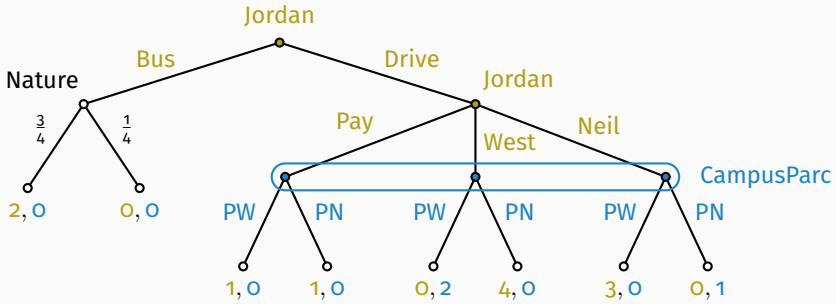
Bonanno Part IV (Chapters 11–13): weak sequential, sequential, and perfect Bayesian equilibrium.

Let’s Work One Out

Gameday Parking, Part 2

Jordan first decides how to get to the game:

- **Bus** (COTA): on time with chance $\frac{3}{4}$ (payoff 2), else late (payoff 0)
- **Drive**: then Chapter 6's parking game. CampusParc can't see where Jordan parked



Gameday Parking, Part 2

1. List Jordan's pure strategies. How many numbers for a mixed strategy? A behavioral one?

2. What is Jordan's expected payoff from Bus?

3. Solve the parking subgame (recall Chapter 6). Payoffs?

4. Find the SPNE. Write it in behavioral strategies

5. How reliable must the bus be before Jordan takes it?

What Have We Learned, Charlie Brown?

- Nature in a tree? Replace the lottery with its **expected utility**
- **Behavioral strategy**: one probability distribution per information set
  - Mixed: spin one wheel over whole plans. Behavioral: flip as you go
- **Kuhn (1953)**: with perfect recall, behavioral $=$ mixed
- **Selten (1965)**: with cardinal payoffs, a SPNE always exists
  - Solve subgames with mixed NE when there's no pure one
- SPNE can still allow silly choices at unreached infosets (Selten's horse)

Reading & Practice

- Bonanno, *Game Theory*, Chapter 7
  - §7.1: read for behavioral strategies and Kuhn's theorem. Skim the formal "game with cardinal payoffs" part
  - §7.2–7.3: read carefully
- Exercises: §7.4.1 (7.1, 7.2, 7.3), §7.4.2 (7.5, 7.7)
  - 7.4 and 7.6 review Chapter 5's utility calibration
  - 7.8 is harder. 7.9 (the mugger) is a challenge
- Solutions in §7.5. Try them first